

Digitalisation technologies for integration span the full maturity range, from deployed at scale to research and prototype

	Sector coupling linking electricity with heat, transport and industry	Flexibility provision shifting demand and supply in time	Real-time balancing matching supply and demand second by second	Forecasting and anticipatory planning predicting output and demand ahead of time
Sensing and data acquisition <i>(internet of things, sensor networks)</i>	● Demonstrated in pilots (Rodrigues et al., 2025; Koirala et al., 2024)	● Deployed at scale (Pal et al., 2021; IEA, 2023)	● Deployed at scale (Pal et al., 2021; Borgaonkar et al., 2021; Jorgensen and Ma, 2025)	● Deployed at scale (IEA, 2023; Kiasari et al., 2024)
Analytics and optimisation <i>(artificial intelligence, machine learning)</i>	○ Research and prototype (Pandey et al., 2023)	● Demonstrated in pilots (Pandey et al., 2023; Henao et al., 2025)	● Demonstrated in pilots (Pandey et al., 2023; Henao et al., 2025)	● Deployed at scale (Pandey et al., 2023; Price et al., 2025)
Simulation and planning <i>(digital twins)</i>	● Demonstrated in pilots (Pálenský et al., 2022; Koirala et al., 2024)	● Demonstrated in pilots (Pálenský et al., 2022)	○ Research and prototype (Aghazadeh Ardebili et al., 2024; Thwe et al., 2025)	● Demonstrated in pilots (Aghazadeh Ardebili et al., 2024; Khan et al., 2025)
Coordination and control <i>(platforms, communication networks)</i>	● Demonstrated in pilots (Rodrigues et al., 2025; Neaimeh et al., 2020)	● Deployed at scale (Rodrigues et al., 2025)	● Deployed at scale (Sospiro et al., 2021; Boeding et al., 2024)	● Demonstrated in pilots (PJM, 2025; Cheng et al., 2025)